

- Telecom and aerospace segments will drive the growth of RF communication testers at a CAGR of 10.5 per cent in FY17-FY22.

- The drive test solution market will decline gradually because of higher automation in industries that make error prediction easier.

- Demand for EMC is going to increase from the aerospace and defence sectors as well as new emerging technologies in industrial and automotive sectors.

The higher education sector has witnessed tremendous growth with more than 36,000 higher educational institutes in India. By 2030, it is estimated that higher education system in India would be among the top five in the world in terms of research output with an annual R&D spend of US\$ 140 billion. Demand for low-range T&M products has been increasing among educational institutes and research institutes.

Key market regions

Major application segments for T&M products are in southern and western India. Major industries located in the western region include engineering goods, electricals and electronics. Key markets for T&M products in the western region are as follows:

- Mumbai: navy, aerospace and defence, manufacturing, industrial, education and research, communications, consumer electronics
- Pune: industrial, education and research

Emerging trends to shape up the market

- Automotive (connected cars, vehicle electrification), consumer electronics (smart devices, smart homes) and communications (4G, 5G) segments will drive the future growth of T&M.
- Emerging technologies such as the Internet of Things (IoT), millimetre-wave (mm wave) frequencies and 5G propose significant changes in requirements for test equipment, thus opening new opportunities for T&M market players. Major T&M players have already started offering 5G wireless test solutions to commercial researchers to assist them in the development and verification of new-generation designs.
- Demand for customised equipment with multiple functionalities is a prevailing trend. Customer centricity and continuous innovation will be high on the radar of T&M players to survive in this competitive market. T&M players will focus on continuous innovation, offering unique features like touchscreen facility, Android support, ultra-portability and so on.
- Capex to opex model is gaining momentum; majorly, telecom companies are moving towards rental model to save on capex.

- Nashik: industrial

Southern region has the following markets for T&M products:

- Hyderabad: defence (DRDO RCI labs), industrial, education and research
- Visakhapatnam: navy, with major focus on light engineering, naval base
- Bengaluru: aerospace and defence, consumer electronics, education and research, industrial, communications, with the presence of public sector units such as Hindustan Aeronautics, Hindustan Machine Tools and Bharat Electronics

- Chennai: navy, aerospace and defence, education and research

Major manufacturing sectors in the northern region include electrical and electronics, and plant and machinery, with key focus on Delhi-NCR (industrial, automotive, communications,

consumer electronics, aerospace and defence, education and research) as T&M market.

Engineering and electrical industries are prominent in the eastern region with key focus on Kolkata and nearby regions (industrial, education and research) as T&M market.

Focus on regional presence and tieups with local distributors who have good relations with customers and can be a front sales will lead to better results.

Moving forward

Along with huge opportunities, the Indian market also offers the following challenges to T&M players:

- Less focus on quality
- Highly price-sensitive nature
- No major incentive/subsidy from the government for capital investments in T&M
- No major incentives for exporters of T&M equipment

Therefore Indian T&M players need to focus on the right strategies to grow. For example, continuous innovation and focus on R&D are critical to sustain in the T&M market. Companies offering high-accuracy instruments with compact and innovative features will get a competitive advantage. However, customer awareness generation through regular product demonstrations and on-time aftersales service are must-to-do for any T&M market player. Moreover, T&M companies need to leverage government initiatives like Make in India, Digital India and Smart Cities. High focus on government tenders and maintaining continuous relations with government officials will help T&M players in increasing market share. **EFY**

Experts' view

➤ "With the recent government mandate to have compulsory TEC testing of wireless products, we see a double-digit growth coming up for our products and systems catering to this testing."

—**Nasser Jariwala, director (sales and operations) - T&M, Rohde & Schwarz India**

➤ "Dearth of IoT test equipment is a challenge. Customised and affordable test equipment for IoT startups are in demand."

—**S.A. Srinivasa Moorthy, group chief technology officer, Precision Group**

➤ "Rental market is growing, hence, keeping less room for new T&M products. Tough price competition and low customer budget are some challenges in the Indian T&M market."

—**Madhukar Tripathi, head marcom and optical products, Anritsu India Pvt Ltd**

➤ "The government needs to reserve at least 25 per cent of purchase by government organisations and all government PSUs from MSME units that are indigenous manufacturers of T&M equipment. Some of the test equipment are under zero-per cent basic import customs duty. This hampers the level playing field for indigenous manufacturers and provides undue price advantage to MNCs."

—**V. Vijayakumar, general manager - sales and marketing, Qmax Test Equipments Pvt Ltd**